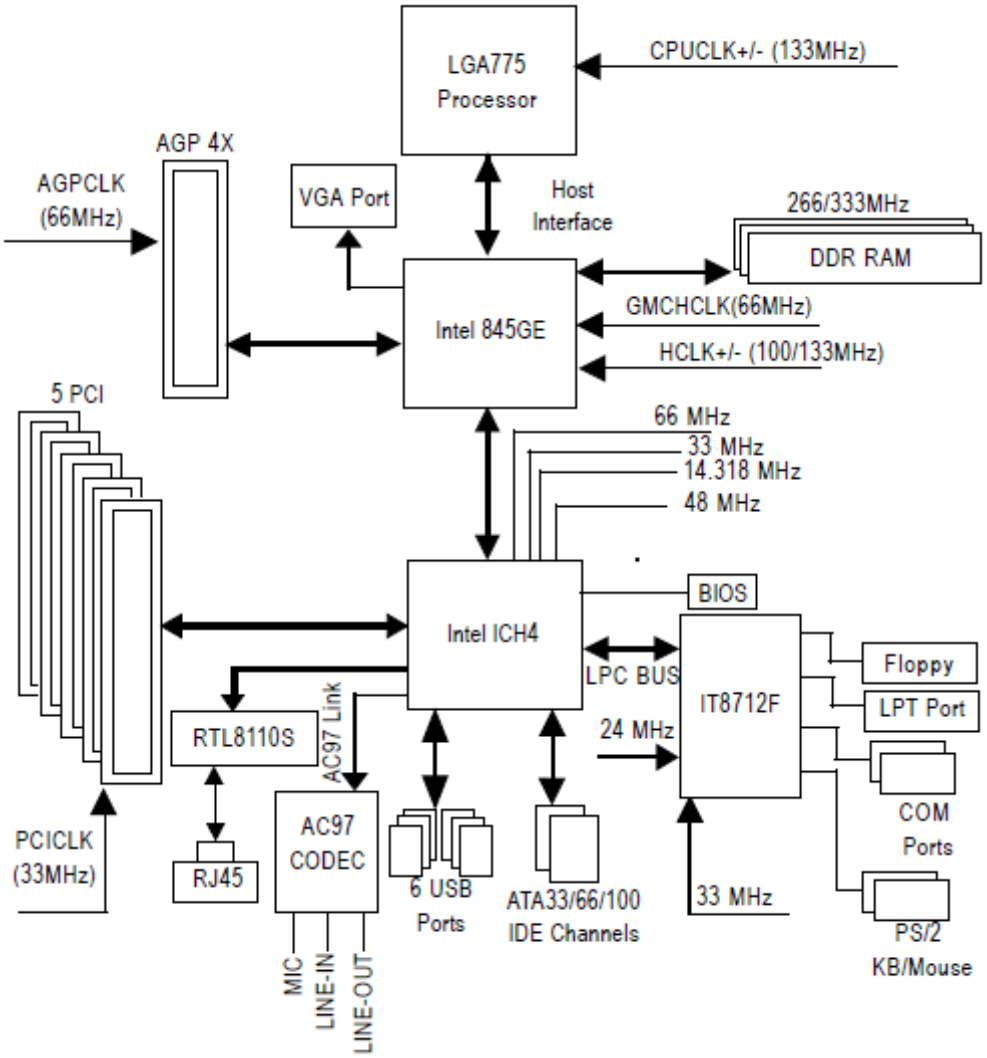
Teoria de plaques base

1.- Amb la marca i el model de placa base cerqueu a Internet els manuals de les plaques base següents i indiqueu TOTS els components següents a TOTES les plaques base:

Ports d'expansió:
PCI, AGP, PCIExpress*1, PCIExpress*16

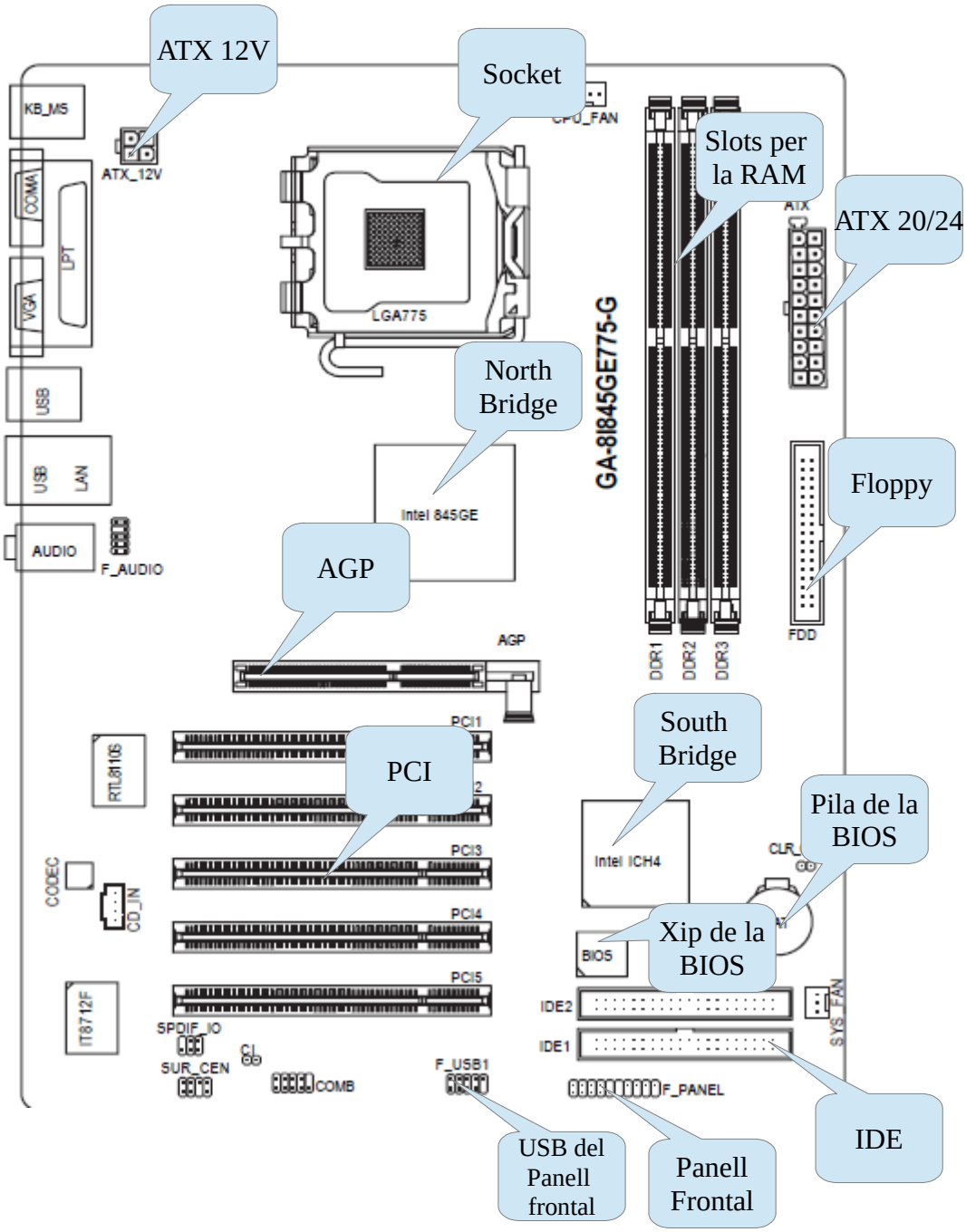
USB's del frontal de caixa
Connector dels controls del frontal de caixa (power led, power button, HDD led i reset button)
Connectors IDE, Floppy i SATA
Connectors de la font d'alimentació ATX 12V i ATX 20/24 pins
Connector del ventilador de la CPU
Connector dels ventiladors addicionals
Pila de la BIOS
Xip de la BIOS
Pont Nord (Northbridge) i pont sud (Southbridge)
Slots per a la Ram
Ports del darrera (Rear panel)
Audio del frontal (auriculars i micròfon)

1.- Placa base gigabyte GA-8I845GE775-G (Intel Pentium4)
FSB (front side bus) 2000-2007

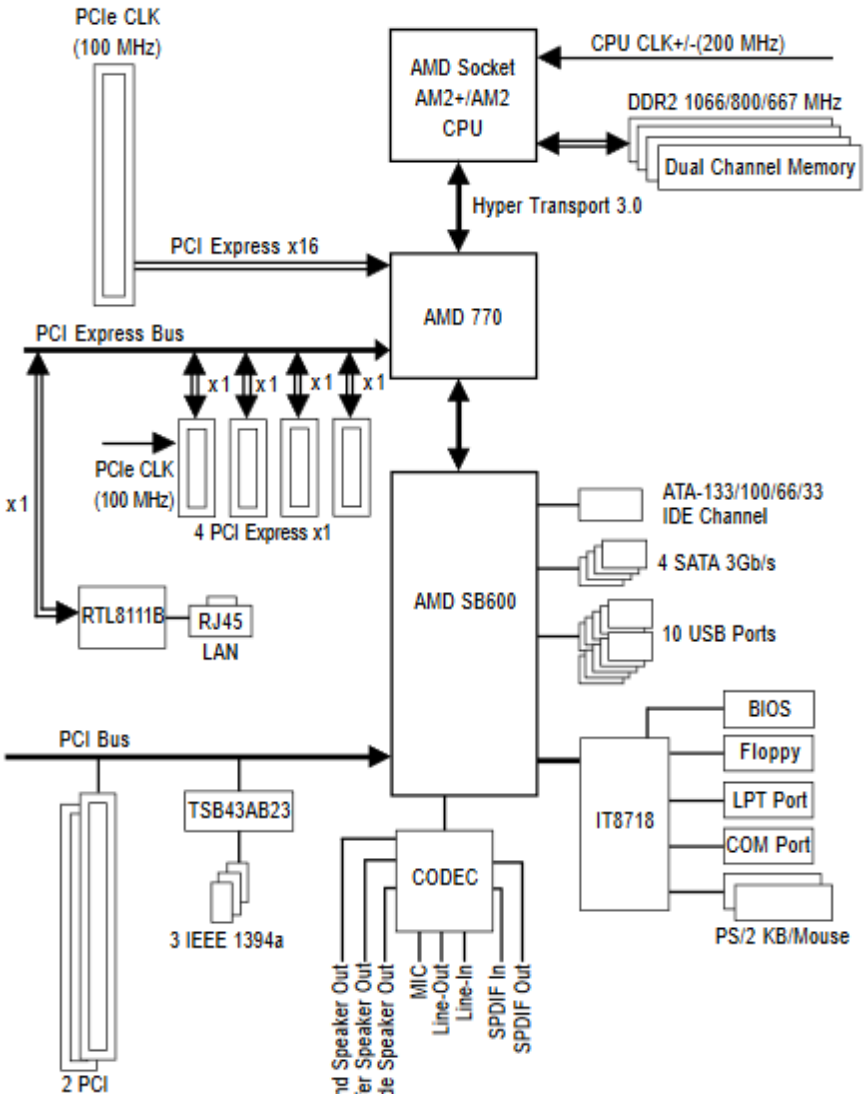


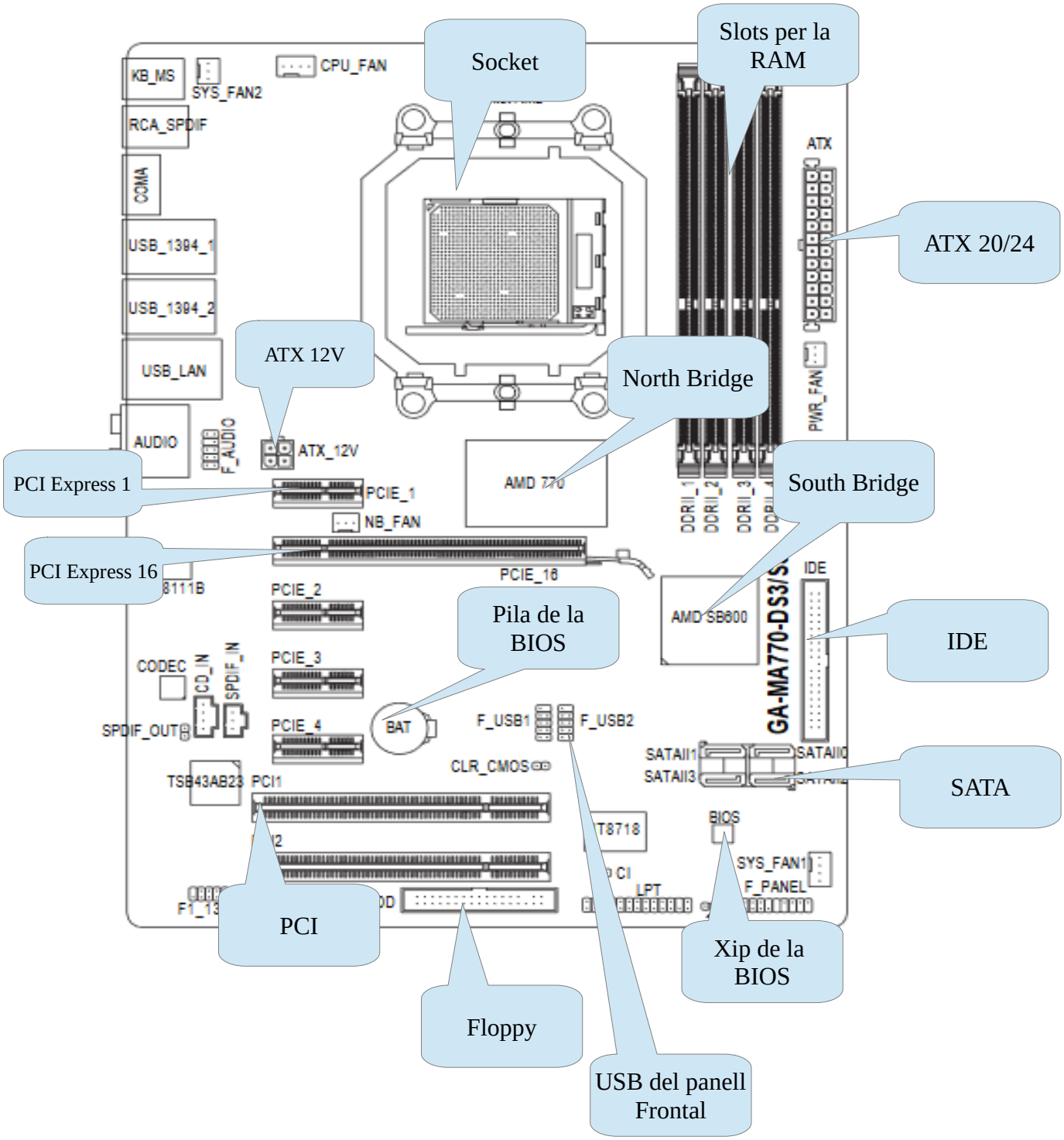


Placa base gigabyte **GA-8I845GE775-G**



2.- GA-MA770-DS3 (AMD) (hyper transport) (2007)

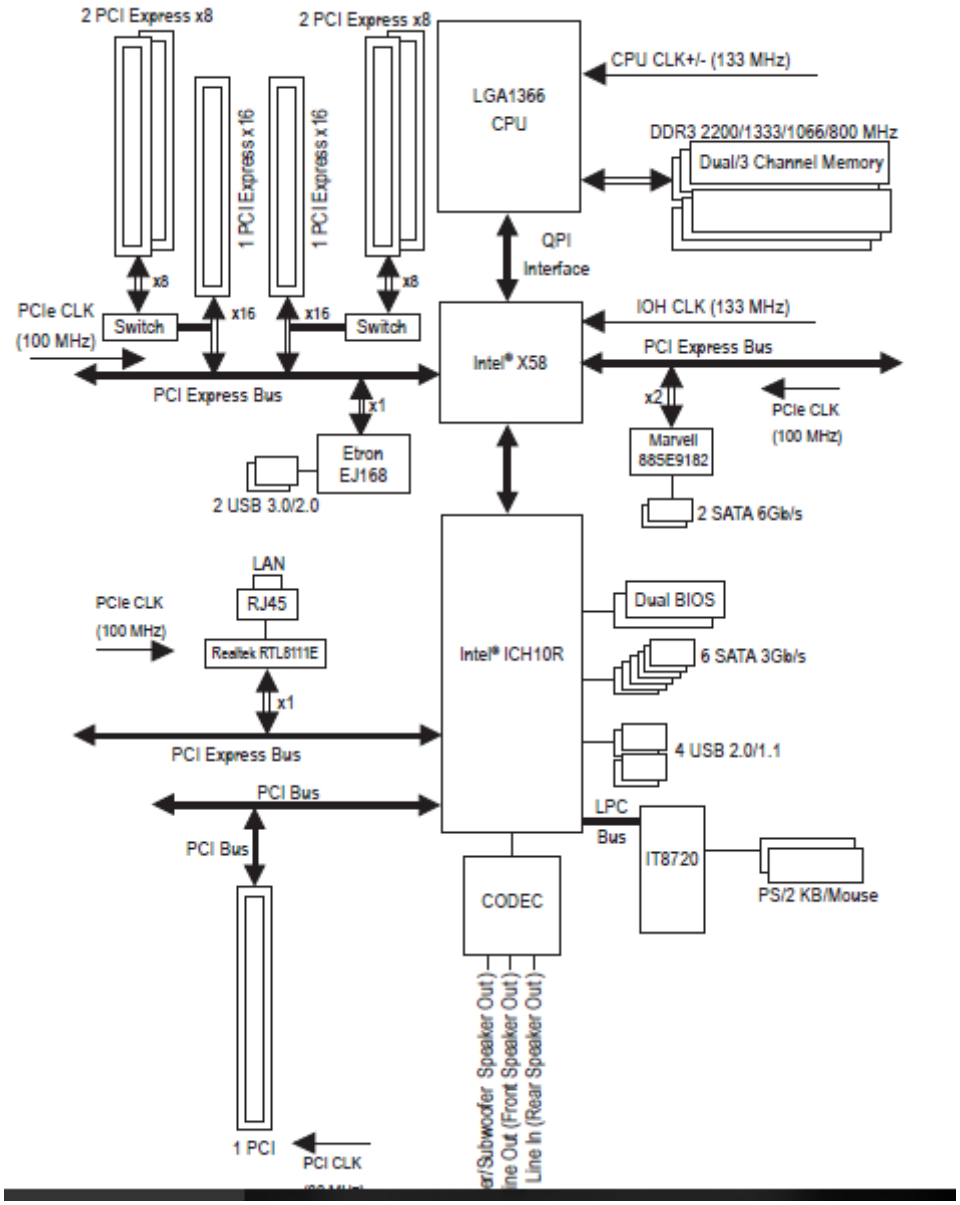


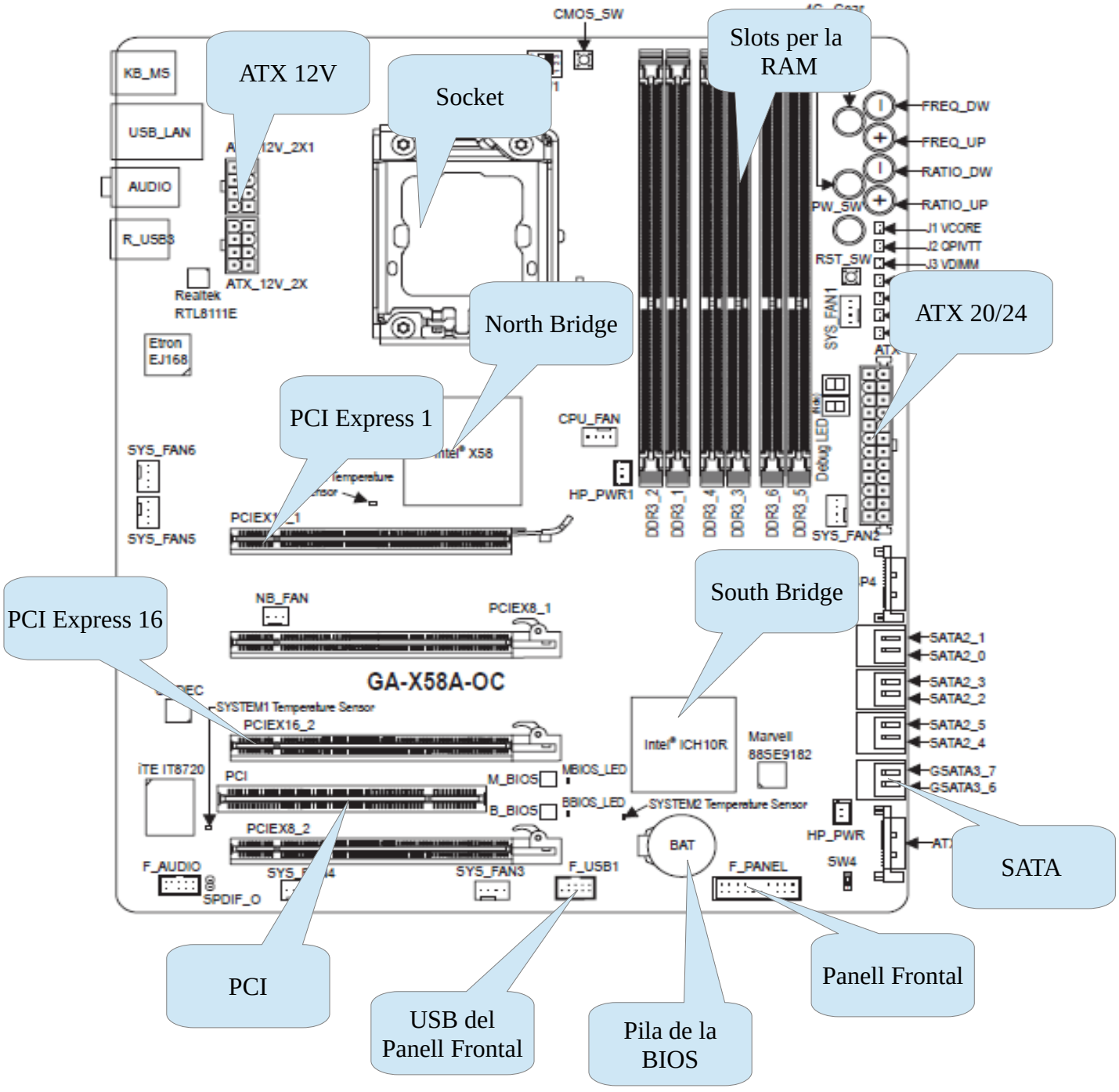

Socket
Slots per la RAM
ATX 20/24
ATX 12V
North Bridge
PCI Express 1
South Bridge
PCI Express 16
Pila de la BIOS
IDE
SATA
PCI
Xip de la BIOS
Floppy
USB del panell Frontal
KB_MS
SYS_FAN2
CPU_FAN
RCA_SPDIF
COMA
USB_1394_1
USB_1394_2
USB_LAN
AUDIO
F_AUDIO
ATX_12V
PCIE_1
NB_FAN
PCIE_16
PCIE_2
PCIE_3
PCIE_4
CODEC
CD_IN
SPDIF_IN
SPDIF_OUT
BAT
F_USB1
F_USB2
CLR_CMOS
TSB43AB23
PCI1
AMD 770
AMD SB600
ATX
PWR_FAN
IDE
SATAII1
SATAII3
BIOS
SYS_FAN1
F_PANEL
LPT
T8718
DDRII_1
DDRII_2
DDRII_3

3.- Placa base gigabyte GA-X58A-OC (Intel)

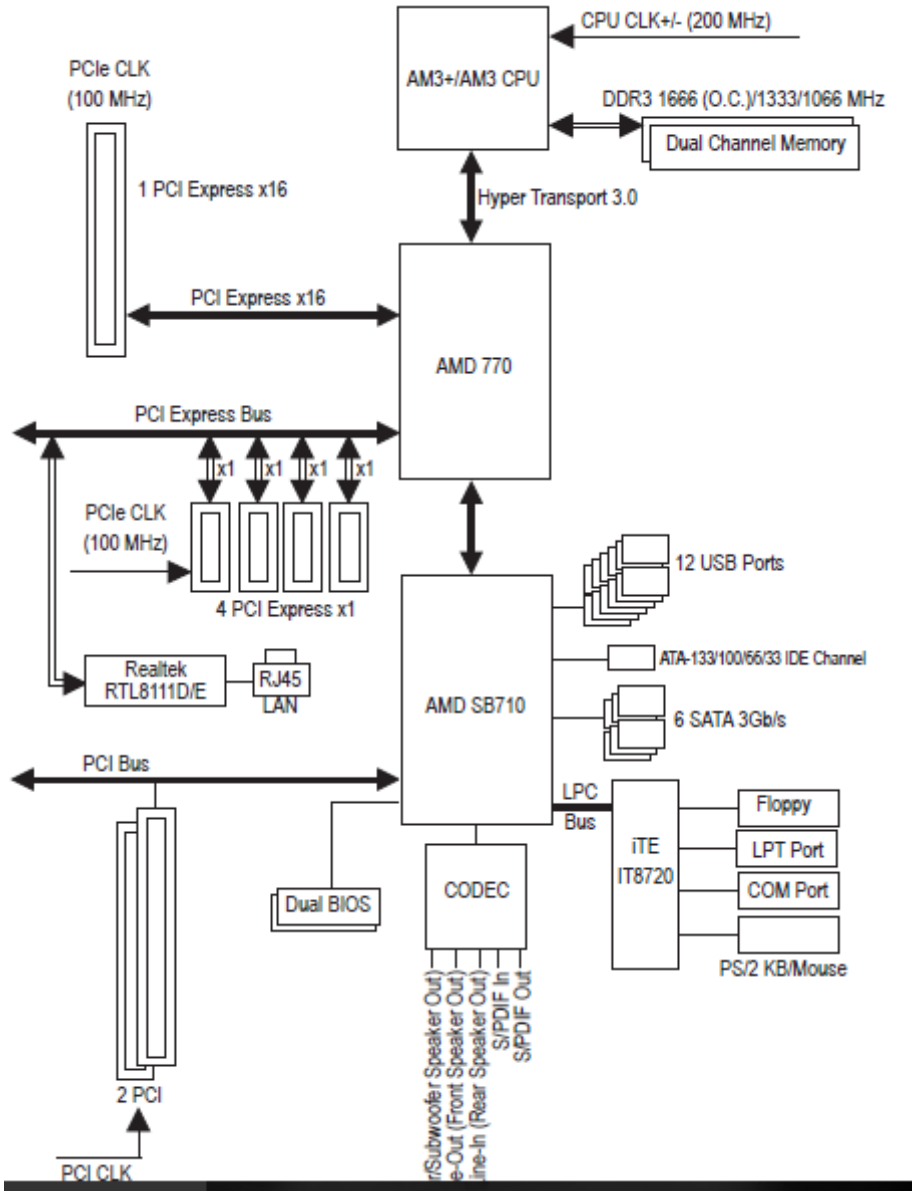
QPI (Quick Path Interconnect) (2008)

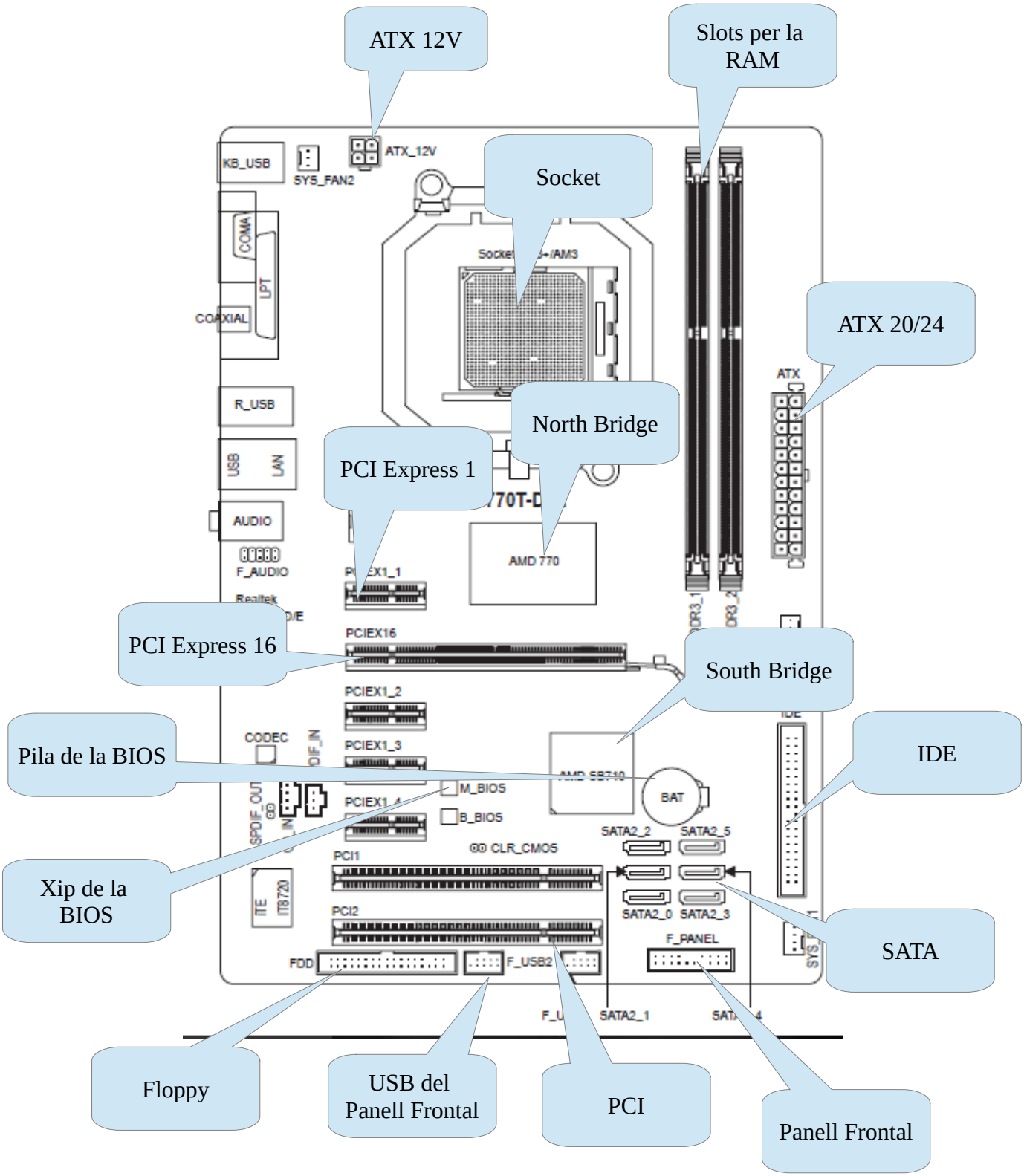
Motherboard Block Diagram




ATX 12V
Socket
Slots per la RAM
North Bridge
ATX 20/24
PCI Express 1
PCI Express 16
South Bridge
SATA
PCI
USB del Panell Frontal
Pila de la BIOS
Panell Frontal
GA-X58A-OC
CMOS_SW
KB_MS
USB_LAN
AUDIO
R_USB3
ATX_12V_2X
Realtek RTL8111E
Etron EJ168
CPU_FAN
HP_PWR1
SYS_FAN6
SYS_FAN5
SYS_FAN1
SYS_FAN2
SYS_FAN3
NB_FAN
PCIEX8_1
PCIEX16_2
PCIEX8_2
PCI
SYSTEM1 Temperature Sensor
SYSTEM2 Temperature Sensor
ITE IT8720
Intel® ICH10R
Marvell 88SE9182
M_BIOS
B_BIOS
MBIOS_LED
BBIOS_LED
BAT
F_AUDIO
SPDIF_O
F_USB1
F_PANEL
HP_PWR
SW4
FREQ_DW
FREQ_UP
RATIO_DW
RATIO_UP
PW_SW
RST_SW
J1 VCORE
J2 OPVTT
J3 VDIMM
Debug LED
DDR3_2
DDR3_1
DDR3_4
DDR3_3
DDR3_6
DDR3_5
SATA2_1
SATA2_0
SATA2_3
SATA2_2
SATA2_5
SATA2_4
GSATA3_7
GSATA3_6

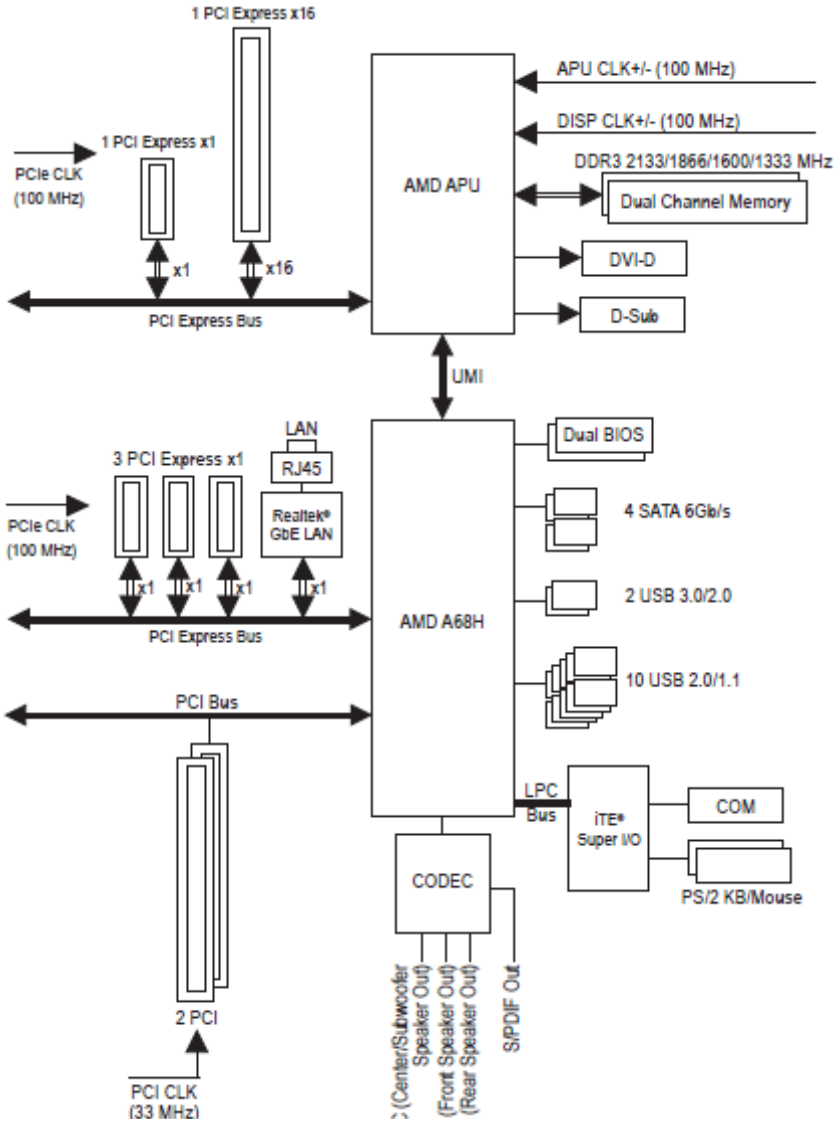
4.- HyperTransport (AMD)
Placa base gigabyte GA-770T-D3L
Motherboard Block Diagram

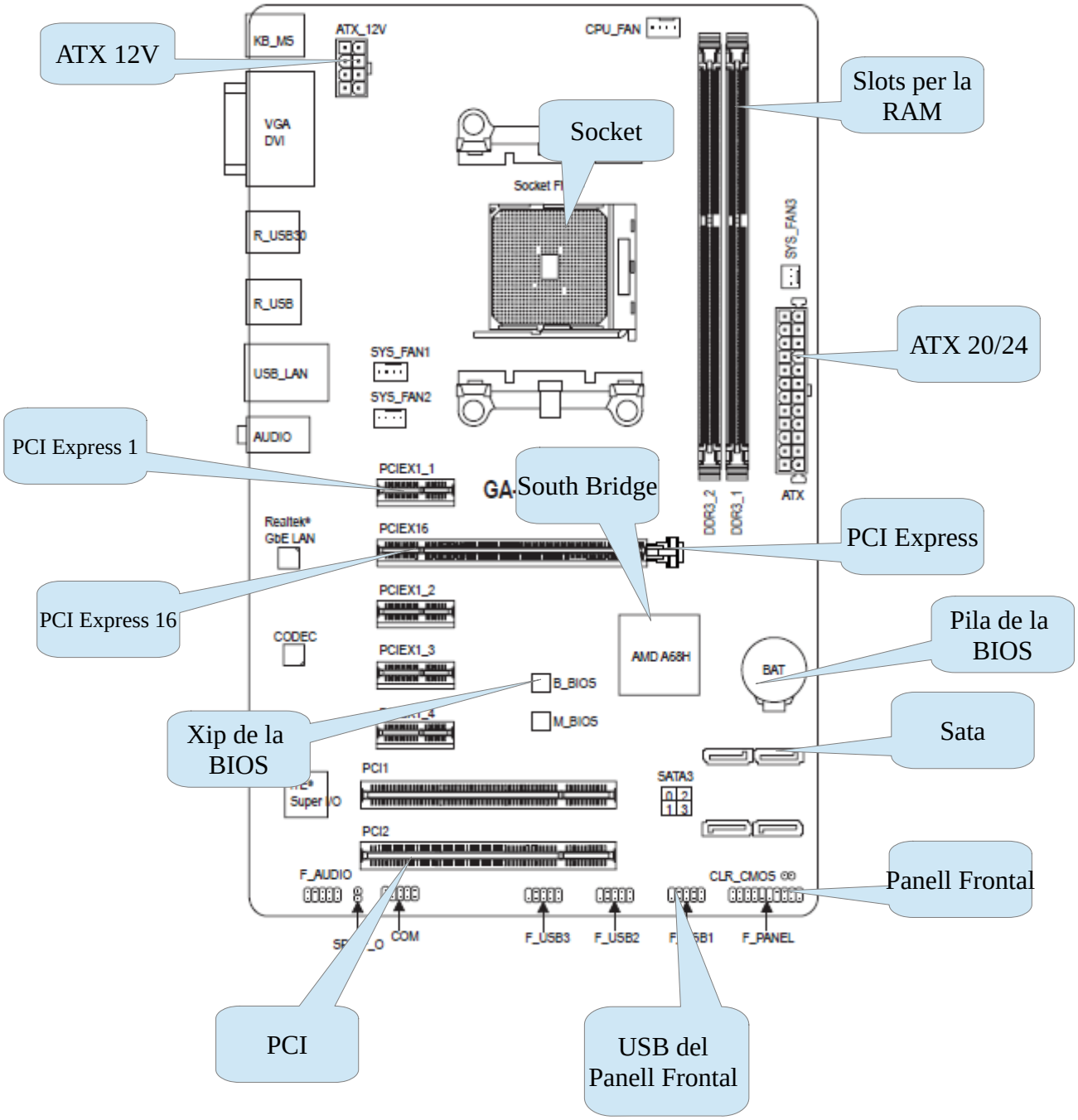



ATX 12V
Slots per la RAM
Socket
ATX 20/24
North Bridge
PCI Express 1
PCI Express 16
South Bridge
Pila de la BIOS
IDE
Xip de la BIOS
SATA
Floppy
USB del Panell Frontal
PCI
Panell Frontal
ATX_12V
SYS_FAN2
KB_USB
COMA
LPT
COAXIAL
R_USB
USB
LAN
AUDIO
F_AUDIO
AMD 770
PCIEX16
PCIEX1_2
PCIEX1_3
PCIEX1_4
CODEC
M_BIOS
B_BIOS
CLR_CMOS
BAT
PCI1
PCI2
FDD
F_USB2
F_PANEL
SATA2_0
SATA2_1
SATA2_2
SATA2_3
SATA2_5
ATX
IDE

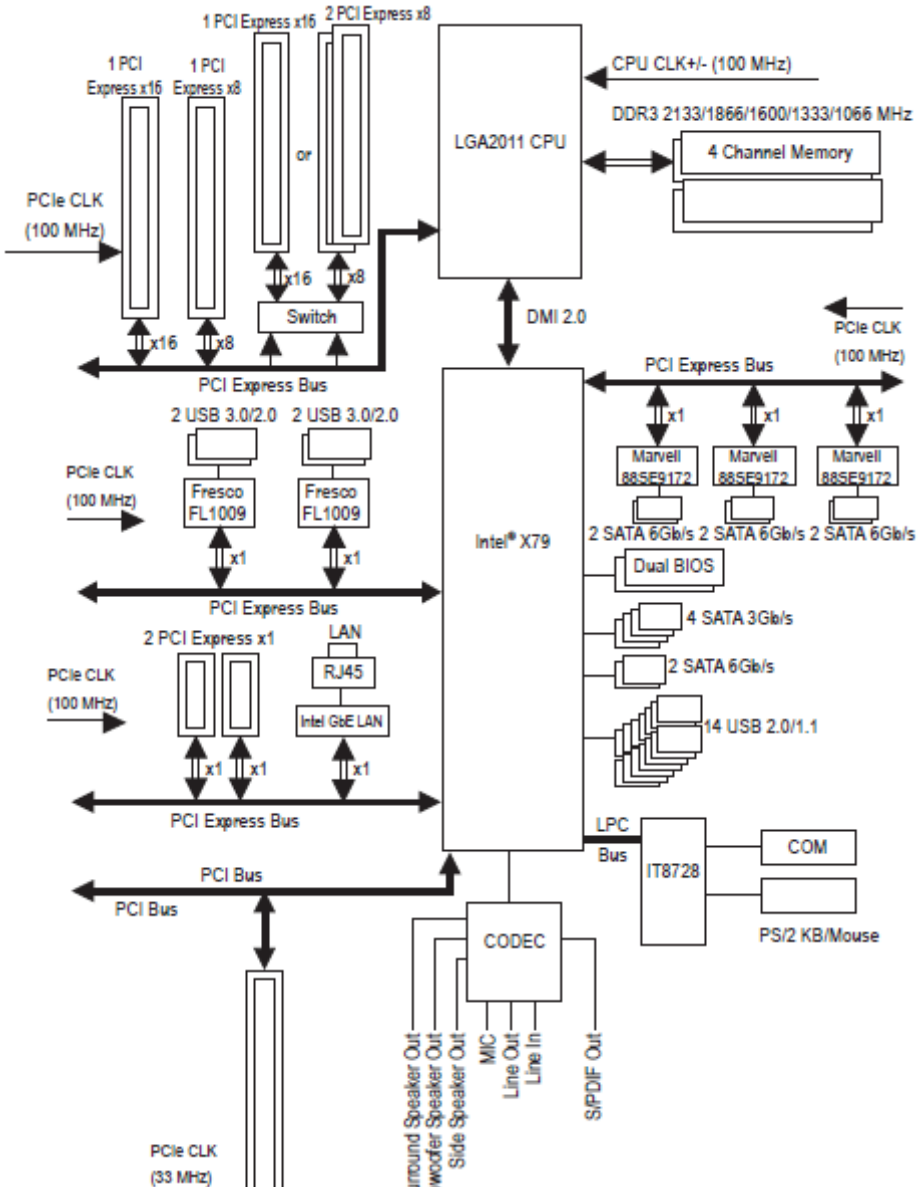
5.- AMD APU

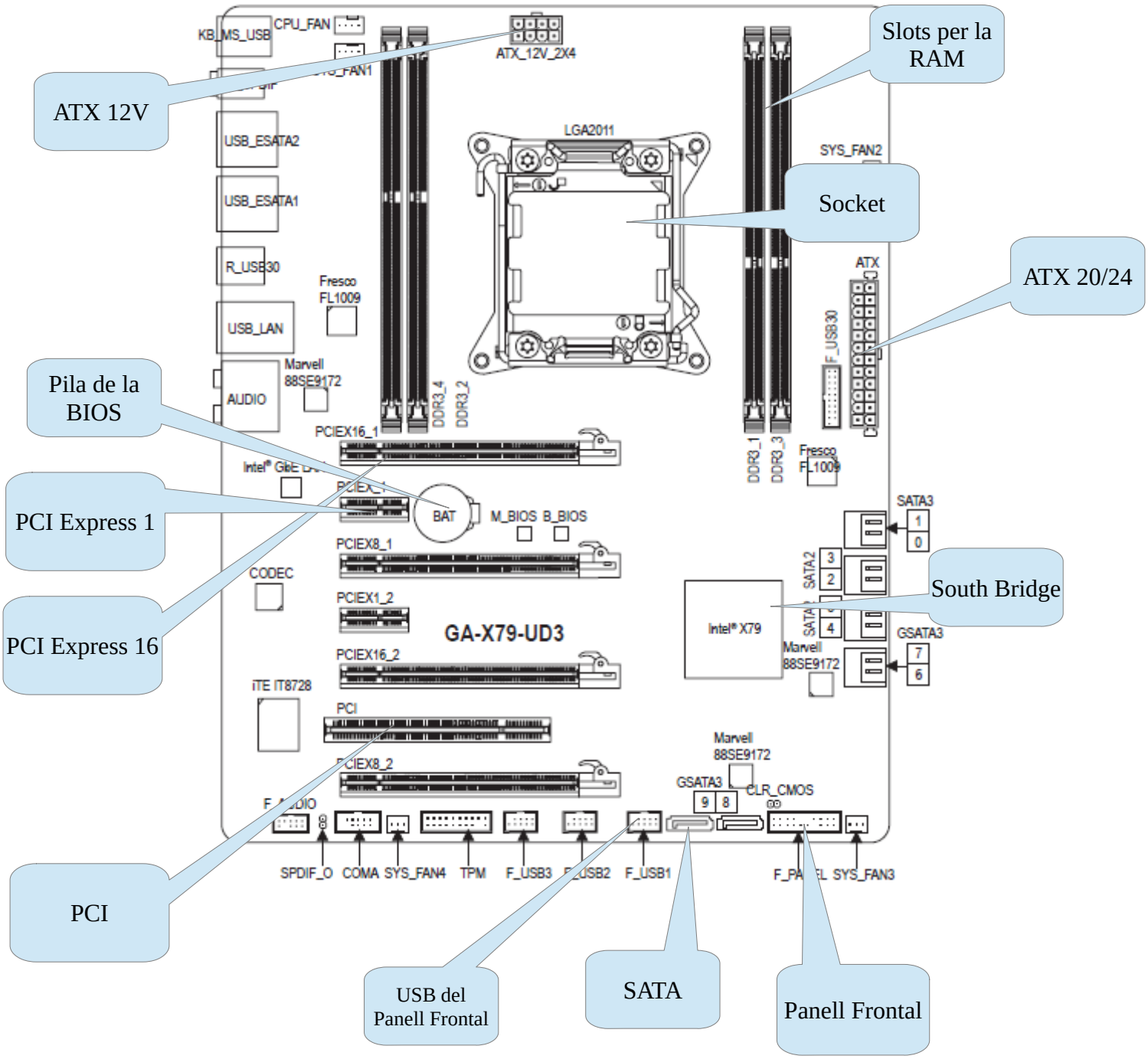
Placa base gigabyte GA-F2A68H-DS3 Motherboard Block Diagram



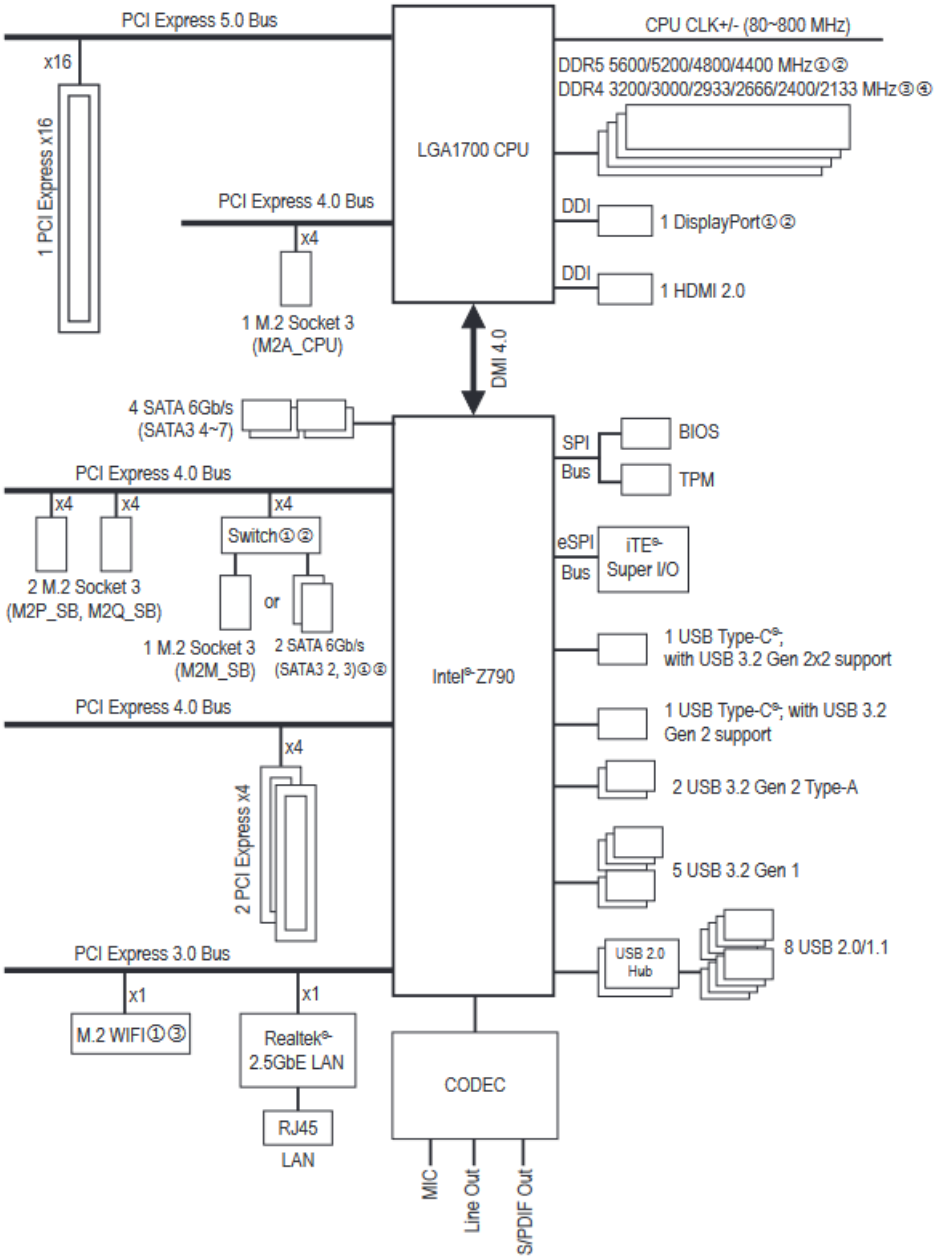

ATX 12V
Slots per la RAM
Socket
ATX 20/24
PCI Express 1
South Bridge
PCI Express
PCI Express 16
Pila de la BIOS
Xip de la BIOS
Sata
Panell Frontal
PCI
USB del Panell Frontal
KB_MS
ATX_12V
CPU_FAN
VGA
DVI
R_USB30
R_USB
USB_LAN
AUDIO
SYS_FAN1
SYS_FAN2
SYS_FAN3
ATX
PCIEX1_1
PCIEX16
PCIEX1_2
PCIEX1_3
Realtek
GbE LAN
CODEC
AMD A68H
BAT
B_BIOS
M_BIOS
PCI1
PCI2
SATA3
F_AUDIO
CLR_CMOS
COM
F_USB3
F_USB2
F_PANEL

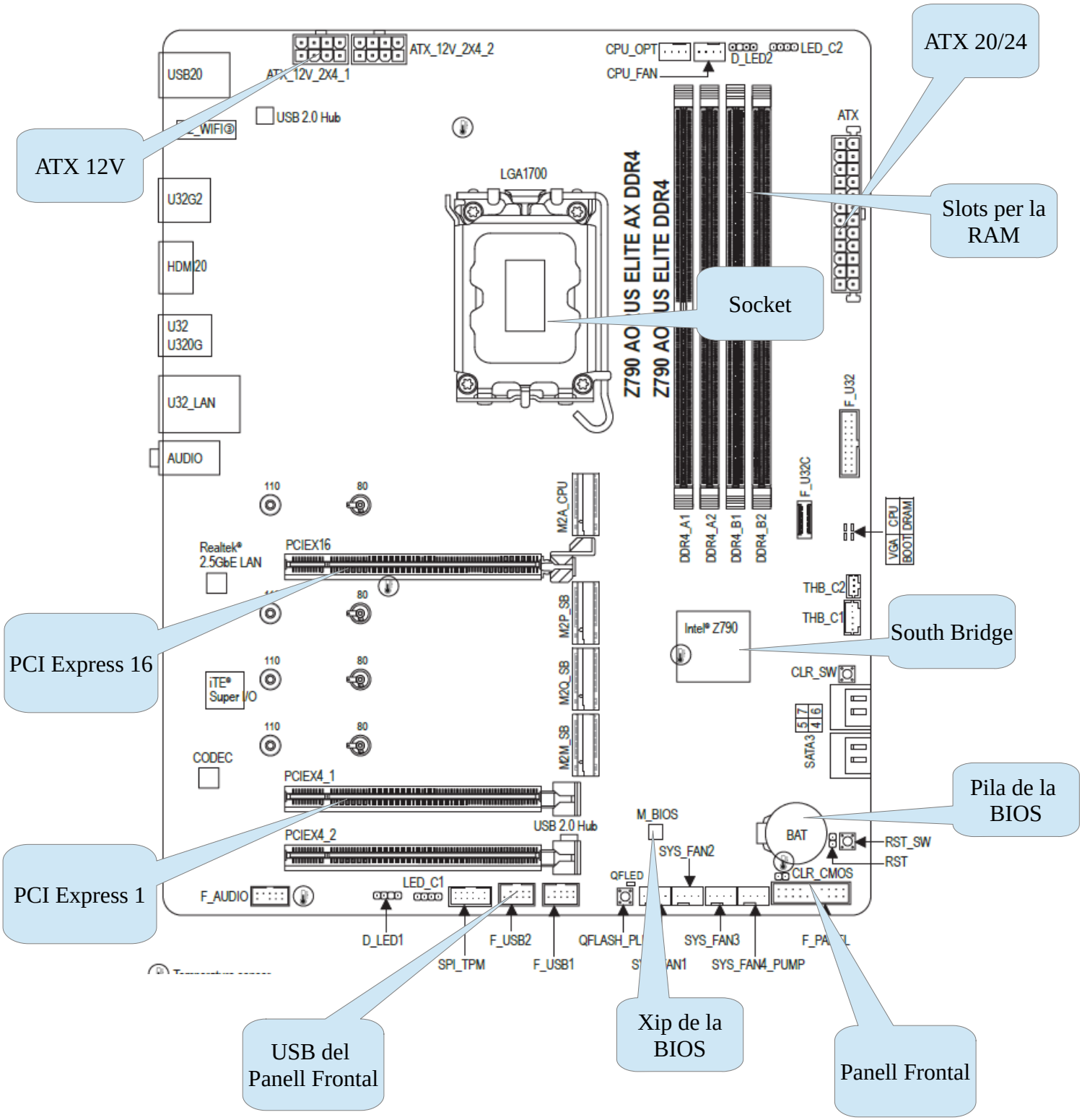
6.- Intel i7 amb ddr3:
Placa base gigabyte GA-X79-UD3
Motherboard Block Diagram



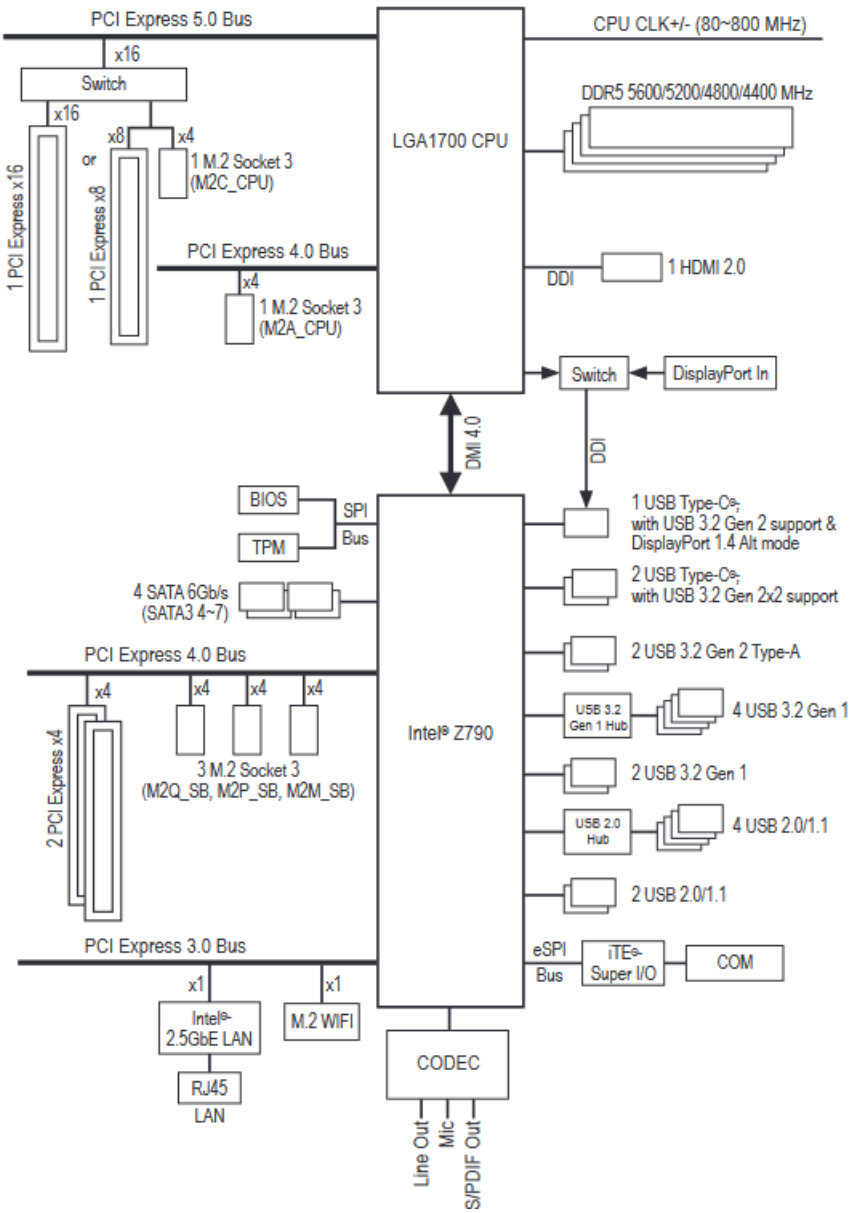

ATX 12V
Slots per la RAM
Socket
ATX 20/24
Pila de la BIOS
PCI Express 1
PCI Express 16
South Bridge
PCI
USB del Panell Frontal
SATA
Panell Frontal
GA-X79-UD3
LGA2011
ATX_12V_2X4
CPU_FAN
KB_MS_USB
USB_ESATA2
USB_ESATA1
R_USB30
USB_LAN
AUDIO
Fresco FL1009
Marvell 88SE9172
DDR3_4
DDR3_2
DDR3_1
DDR3_3
SYS_FAN2
ATX
F_USB30
PCIEX16_1
PCIEX8_1
PCIEX1_2
PCIEX16_2
PCIEX8_2
PCI
BAT
M_BIOS
B_BIOS
CODEC
iTE IT8728
Intel® X79
SATA3
GSATA3
CLR_CMOS
SPDIF_O
COMA
SYS_FAN4
TPM
F_USB3
F_USB2
F_USB1
SYS_FAN3

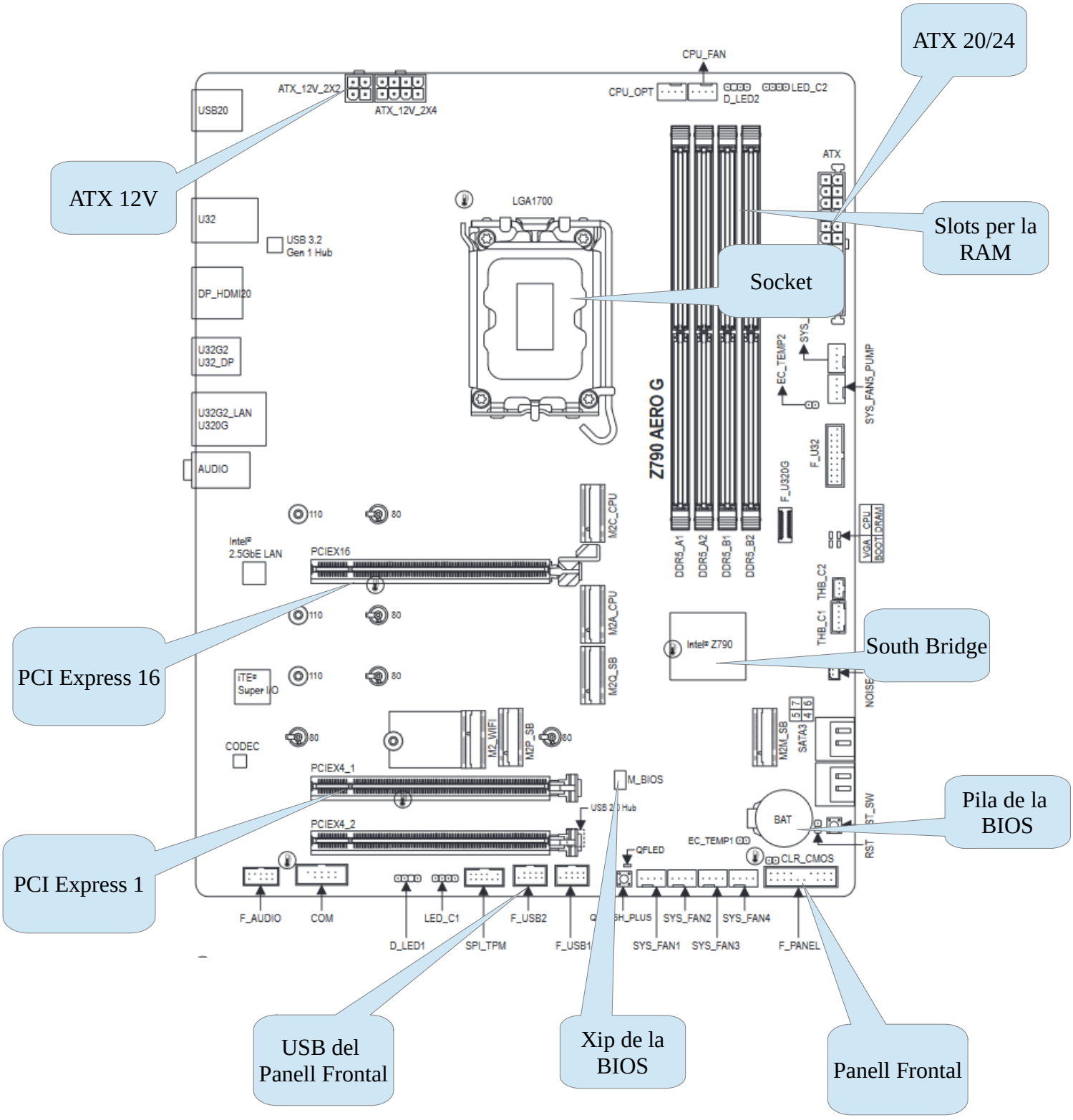
7.- Intel (DDR4)


PCI Express 5.0 Bus
x16
1 PCI Express x16
LGA1700 CPU
CPU CLK+/- (80~800 MHz)
DDR5 5600/5200/4800/4400 MHz①②
DDR4 3200/3000/2933/2666/2400/2133 MHz③④
PCI Express 4.0 Bus
x4
1 M.2 Socket 3
(M2A_CPU)
DDI
1 DisplayPort①②
DDI
1 HDMI 2.0
DMI 4.0
4 SATA 6Gb/s
(SATA3 4~7)
PCI Express 4.0 Bus
x4
x4
x4
Switch①②
2 M.2 Socket 3
(M2P_SB, M2Q_SB)
or
1 M.2 Socket 3
(M2M_SB)
2 SATA 6Gb/s
(SATA3 2, 3)①④
SPI
Bus
BIOS
TPM
eSPI
Bus
iTE Super I/O
Intel Z790
1 USB Type-C; with USB 3.2 Gen 2x2 support
1 USB Type-C; with USB 3.2 Gen 2 support
2 USB 3.2 Gen 2 Type-A
5 USB 3.2 Gen 1
8 USB 2.0/1.1
USB 2.0 Hub
PCI Express 4.0 Bus
x4
2 PCI Express x4
PCI Express 3.0 Bus
x1
x1
M.2 WIFI①③
Realtek 2.5GbE LAN
RJ45
LAN
CODEC
MIC
Line Out
S/PDIF Out


ATX 20/24
ATX 12V
Slots per la RAM
Socket
PCI Express 16
South Bridge
PCI Express 1
Pila de la BIOS
USB del Panell Frontal
Xip de la BIOS
Panell Frontal
ATX_12V_2X4_2
ATX_12V_2X4_1
CPU_OPT
CPU_FAN
D_LED2
LED_C2
USB20
USB 2.0 Hub
LGA1700
ATX
Z790 AORUS ELITE AX DDR4
Z790 AORUS ELITE DDR4
U32G2
HDMI20
U32 U320G
U32_LAN
AUDIO
F_U32
F_U32C
VGA CPU BOOT DRAM
DDR4_A1
DDR4_A2
DDR4_B1
DDR4_B2
M2A_CPU
M2P_SB
M2Q_SB
M2M_SB
Realtek® 2.5GbE LAN
PCIEX16
PCIEX4_1
PCIEX4_2
iTE® Super I/O
CODEC
Intel® Z790
THB_C2
THB_C1
CLR_SW
SATA3
BAT
RST_SW
RST
CLR_CMOS
M_BIOS
SYS_FAN2
QFLED
F_AUDIO
LED_C1
D_LED1
SPI_TPM
F_USB2
F_USB1
QFLASH_PL
SYS_FAN3
SYS_FAN4_PUMP

8.- Intel Actual (DDR5)




ATX 20/24
ATX 12V
Slots per la RAM
Socket
PCI Express 16
South Bridge
PCI Express 1
Pila de la BIOS
USB del Panell Frontal
Xip de la BIOS
Panell Frontal
CPU_FAN
ATX_12V_2X2
ATX_12V_2X4
CPU_OPT
D_LED2
LED_C2
ATX
LGA1700
USB20
U32
USB 3.2 Gen 1 Hub
DP_HDMI20
U32G2 U32_DP
U32G2_LAN U320G
AUDIO
Z790 AERO G
DDR5_A1
DDR5_A2
DDR5_B1
DDR5_B2
EC_TEMP2
SYS_FAN5_PUMP
F_U32
F_U320G
VGA_CPU BOOT DRAM
THB_C1
THB_C2
NOISE
M2C_CPU
M2A_CPU
M2Q_SB
M2M_SB
M2_WIFI
M2P_SB
Intel® 2.5GbE LAN
PCIEX16
PCIEX4_1
PCIEX4_2
iTE® Super I/O
CODEC
Intel® Z790
SATA3
M_BIOS
BAT
CLR_CMOS
EC_TEMP1
QFLED
RST
F_AUDIO
COM
LED_C1
D_LED1
SPI_TPM
F_USB2
F_USB1
SYS_FAN1
SYS_FAN2
SYS_FAN3
SYS_FAN4
F_PANEL